

LINEAR ALGEBRA- 2

DOT PRODUCT &
HYPERPLANE

Recap.

* Machine Learning Context : training classifiers

* New terms

(a) Feature

(b) label

(c) datapoint

(d) Dataset

(e) classifiers

* How to train classifier. "optimisation" (last 5 classes).

* Lines : $y = mx + c$
 $\xrightarrow{\text{slope}}$ $\xrightarrow{\text{y intercept}}$

<https://www.desmos.com/calculator>

<https://www.geogebra.org/3d?lang=en>

$$\underbrace{w_1 x_1 + w_2 x_2 + w_0}_{\text{parameters}} = 0$$

features.

find parameters manually

Class 1

w	H	Width
x_1	 y_1	
62	71	30
66	88	40
80	55	50
99	75	22
98	81	45
79	94	.
51	61	.
57	90	.
82	67	.
64	93	.

Class 2

w	H	Width
x_2	 y_2	
33	27	29
18	12	29
47	40	54
23	19	59
24	11	60
46	13	.
10	19	.
22	26	.
24	43	.

$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

$$1 x_1 + 1 x_2 + -98 = 0$$

$$x_1 + x_2 = 98 \rightarrow \text{line}$$

- Will I be able to separate using this line $\rightarrow$ NO
- What should we do? $\rightarrow$ A plane
- What is the eqⁿ of plane!

(2D)

$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

→ General eqⁿ of line
2D Hyperplane

(3D)

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0 = 0 \rightarrow \text{General eqⁿ of a plane}$$

3D Hyperplane

(nD)

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + \dots + w_n x_n + w_0 = 0$$

↳ General eqⁿ of
nD Hyperplane.

Linear Algebra - Vectors

list, tuple, numpy arrays, pandas series.

What is a vector?

- * Magnitude & Direction $\rightarrow$ physics
- * Collection of numbers.

$\vec{x}$ $\swarrow$ standard vector.

How do we represent vector?

If $\vec{x}$ has d dimensions we can write

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_d \end{bmatrix}$$

Column Vector

$$x \in \mathbb{R}^d$$

$\mathbb{R} \rightarrow$ Sets of all the real numbers

$\in \rightarrow$ belongs to

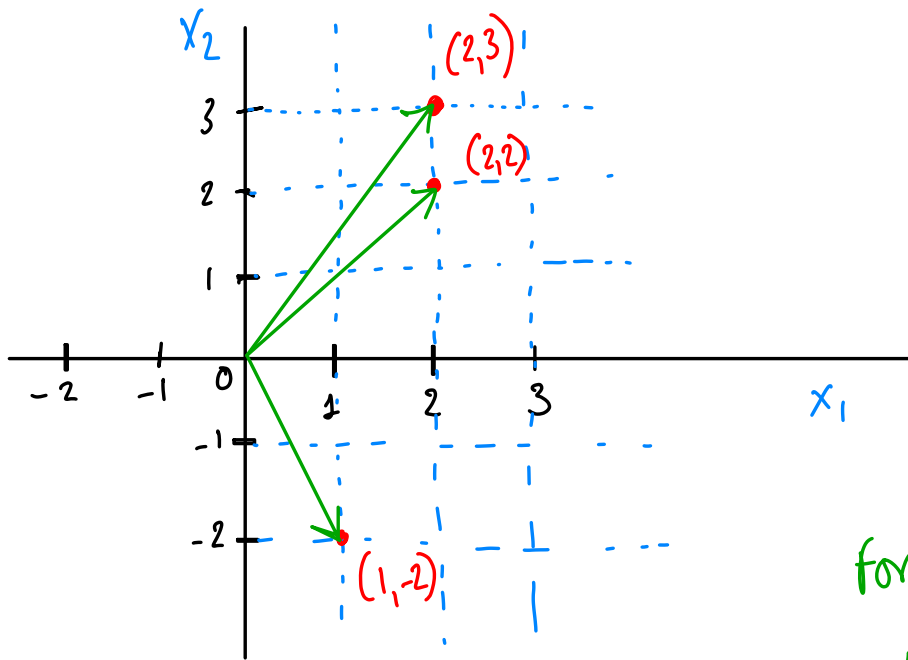
Linear Algebra - Vectors

$$\vec{x} = [x_1 \ x_2 \ x_3 \ \dots \ x_d] \text{ row vector.}$$

① How to visualise Vector?

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\vec{x}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \vec{x}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad \vec{x}_3 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$



* What is the magnitude of these vector?

It is the length of line joining origin to the point.

for 2D vectors magnitude is given by $\sqrt{x_1^2 + x_2^2}$

Linear Algebra - Vectors -

* How to get magnitude of vector in d-dimension?

$$\text{For 2D, } \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\text{magnitude}(\vec{x}) = \sqrt{x_1^2 + x_2^2}$$

$$\text{For 3D, } \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\text{magnitude}(\vec{x}) = \sqrt{x_1^2 + x_2^2 + x_3^2}$$

$$\text{For dD, } \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_d \end{bmatrix}$$

$$\text{magnitude}(\vec{x}) = \sqrt{x_1^2 + x_2^2 + x_3^2 + \dots + x_d^2}$$

Linear Algebra - Vectors

* Norm of a vector is its length.

* We can write norm of a vector using symbol. $\| \|$

norm of a vector $\vec{x}$ $\rightarrow \| \vec{x} \|$

* L_2 Norm $\rightarrow$ The formula for L_2 Norm of a vector ($\vec{x}$)

$$\| \vec{x} \|_2 = \sqrt{x_1^2 + x_2^2 + x_3^2 \dots x_d^2}$$

similar to euclidean distance

* L_1 Norm $\rightarrow$ formula for L_1 Norm of a vector ($\vec{x}$)

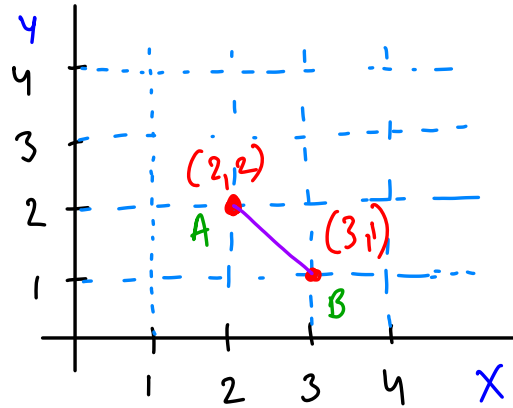
$$\| \vec{x} \|_1 = |x_1| + |x_2| + |x_3| \dots + |x_d|$$

similar to Manhattan.

Linear Algebra - Vectors -

(1) Euclidean Distance
shortest path b/w A & B

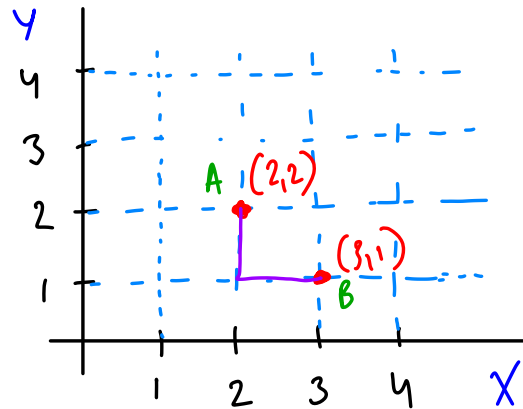
$$\sqrt{(3-2)^2 + (1-2)^2} \approx \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$



(2) Manhattan Distance

The distance by grid shaped
roads on a 2D plane

$$|3-2| + |1-2| \approx |x_2-x_1| + |y_2-y_1|$$



Linear Algebra: Matrix Multiplication

$$A \begin{bmatrix} \text{1} & \text{2} \\ \text{3} & \text{4} \end{bmatrix}_{2 \times 2} \quad B \begin{bmatrix} \text{5} & \text{6} \\ \text{7} & \text{8} \end{bmatrix}_{2 \times 2} = \begin{bmatrix} 1 \times 5 + 2 \times 7 & 1 \times 6 + 2 \times 8 \\ 3 \times 5 + 4 \times 7 & 3 \times 6 + 4 \times 8 \end{bmatrix}$$

Dimension of a matrix: #Rows $\times$ #Columns

* To perform matrix multiplication we have to check

$$A_{m \times n} \quad B_{a \times b} \Rightarrow n = a$$

* Shape of resultant matrix $\Rightarrow$

$$\boxed{[]_{m \times b}}$$

Linear Algebra. Angle b/w 2 Vectors.

$$\bar{x} \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{2 \times 1}$$

$$\bar{y} \begin{bmatrix} 3 \\ 4 \end{bmatrix}_{2 \times 1}$$

if $\bar{x}$ is a vector,
transpose of $\bar{x}$ is $\bar{x}^T$

transpose

$$\begin{bmatrix} 1 & 2 \end{bmatrix}_{1 \times 2}$$

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix}_{2 \times 1}$$

$$= \begin{bmatrix} 1 \times 3 + 2 \times 4 \end{bmatrix}_{1 \times 1}$$

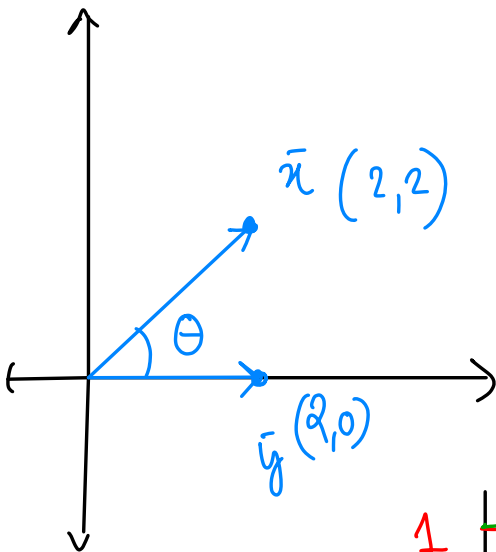
This operation is known as DOT PRODUCT.

$\bar{x}^T \bar{y}$ → Notation we use for dot product.

$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{2 \times 1} y \begin{bmatrix} 3 & 4 \end{bmatrix}_{1 \times 2} = \begin{bmatrix} 1 \times 3 & 1 \times 4 \\ 2 \times 3 & 2 \times 4 \end{bmatrix}_{2 \times 2}$$

$$\bar{x} \bar{y}^T$$

Connection b/w Coordinate Geometry and Linear Algebra.



Θ = Angle b/w vectors $\vec{x}$ & $\vec{y}$

$$\cos(\theta) = \frac{\vec{x}^T \vec{y}}{\|\vec{x}\| \cdot \|\vec{y}\|}$$

$$\cos(\theta) \geq 0$$

+ve

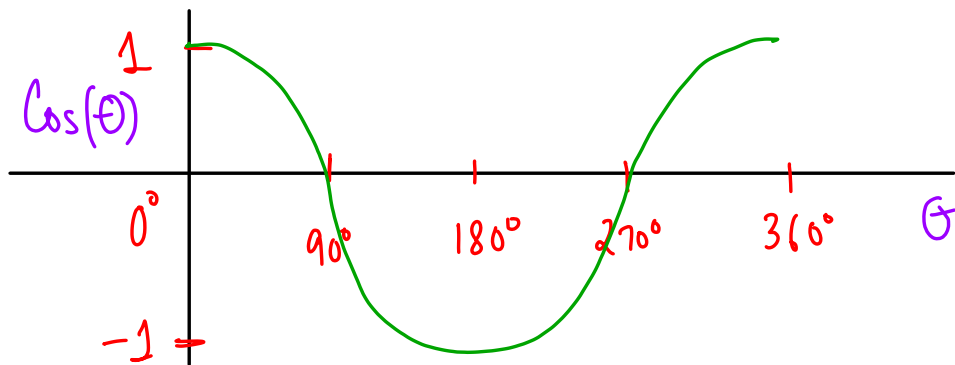
$$0^\circ \leq \theta \leq 90^\circ$$

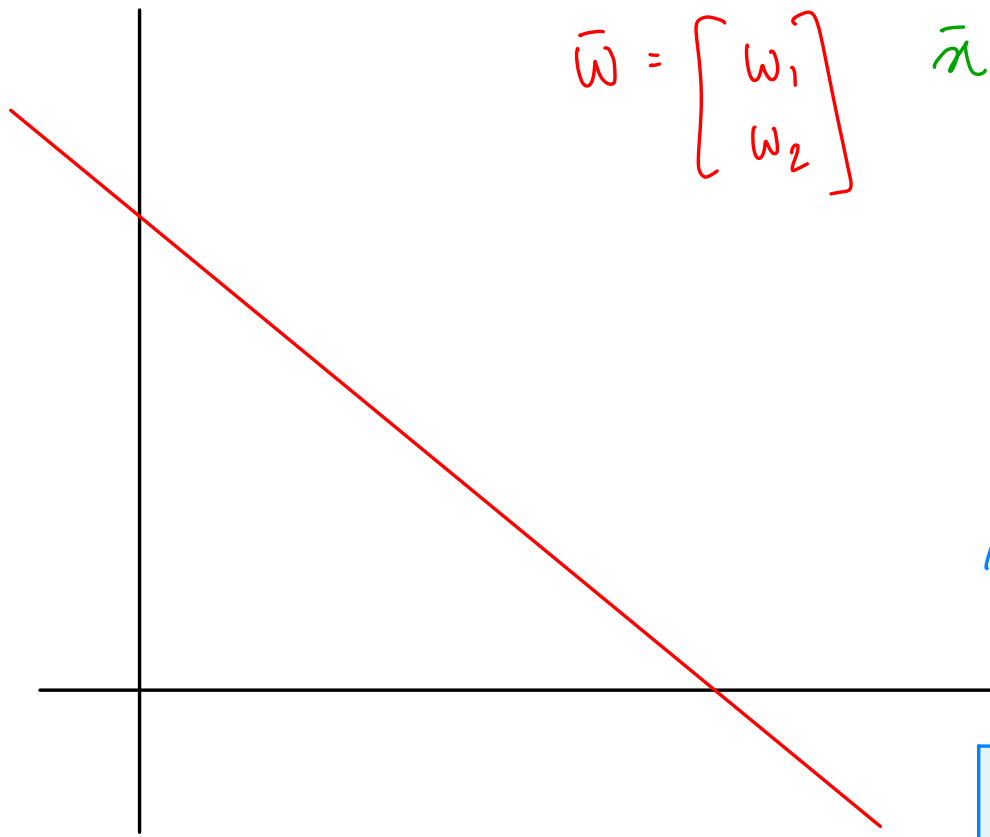
or

$$270^\circ \leq \theta \leq 360^\circ$$

$$\cos(\theta) \leq 0$$

$$90^\circ \leq \theta \leq 270^\circ$$





$$\bar{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\bar{w}^T \bar{x} + w_0 = 0$$

Interpret this eqⁿ
of line using Vector.

$$\text{Eq. } \underbrace{w_1 x_1 + w_2 x_2}_{\bar{w}^T \bar{x}} + w_0 = 0$$

$$\bar{w}^T \bar{x}$$

(d D)

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + \dots + w_d x_d + w_0 = 0$$

Weight
Vector.

$$\bar{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_d \end{bmatrix}$$

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_d \end{bmatrix}$$

→ feature vector

$$\bar{w}^T \bar{x} + \underbrace{w_0}_{\text{bias}} = 0$$

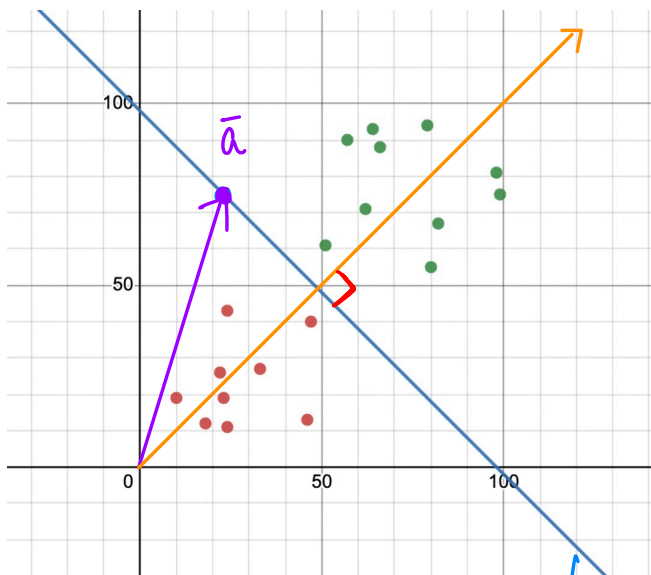
$$\bar{w}^T x + w_0 = 0$$

$$\bar{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ weight vector}$$

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \text{ feature vector}$$

$$w_0 = -98$$

Weight Vector is $\perp$ (perpendicular) to the line.



$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

$\downarrow$ $\downarrow$ $\downarrow$
 1 1 -98

$$x_1 + x_2 - 98 = 0$$

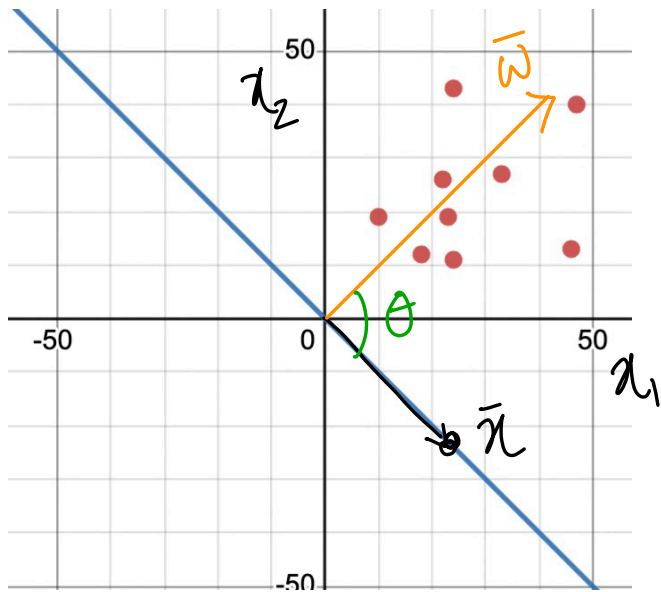
$$\bar{a} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$x + y - 98 = 0$$

$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

$$1x + 1y - 98 = 0$$

$$x + y - 98 = 0$$



$$\bar{w}^T \bar{x} = 0 \quad \{ w_0 = 0 \}$$

$$\cos(\theta) = \frac{\bar{w}^T \bar{x}}{\|\bar{w}\| \cdot \|\bar{x}\|}$$

$$\cos \theta = \frac{0}{\|\bar{x}\| \|\bar{w}\|} \quad \bar{w}^T \bar{x} = 0$$

$$\cos \theta = 0 \rightarrow \theta = 90^\circ$$

	0°	30°	45°	60°	90°
Sin	$\sqrt{\frac{0}{4}} = 0$	$\sqrt{\frac{1}{4}} = \frac{1}{2}$	$\sqrt{\frac{2}{4}} = \frac{1}{\sqrt{2}}$	$\sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$	$\sqrt{\frac{4}{4}} = 1$
cos	1	$\sqrt{3}/2$	$1/\sqrt{2}$	1/2	0
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞
cot	∞	$\sqrt{3}$	1	$1/\sqrt{3}$	0
sec	1	$2/\sqrt{3}$	$\sqrt{2}$	2	∞
cosec	∞	2	$\sqrt{2}$	$2/\sqrt{3}$	1